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Website:

<https://pudzianpmnis.netlify.app/>

Github repository:

<https://github.com/DanielGGMU/mPMNIS>

Roamly Project Alignment and Prototype Overview

We designed the Roamly prototype to directly address the administrative and linguistic overload we identified in our UX research. As students ourselves, we wanted to ensure that our solution genuinely helps new exchange students like our persona Alex settle into their new host city. Here is a comprehensive overview of how our HTML prototype aligns with our established AI design policies, user needs, and mental models.

1. Evidence of User Need and Persona Alignment

Our landing page perfectly introduces our target persona, Alex the Overwhelmed Explorer. As outlined in our newest research, Alex is a 21 year old third year bachelor student dealing with an immense mental load from administrative procedures and a foreign language. The app prototype defaults to the profile of Alex during their first few days, successfully targeting the critical arrival window defined for User Group A. This ensures we solve the pervasive problem of repetitive document translation and event searching.

2. Workflow Transformation and AI Suitability

We designed the app to shift users from a variable novice workflow, where they constantly switch between various maps and translation apps, to a uniform expert workflow within a centralized platform.

- **Language and Administration:** In the Paperwork tab, users can upload local forms. Instead of relying on scattered tools, the AI translates the official housing documents into their native language.
- **Personalization and Smart Discovery:** To bypass the negative mental model Alex has for generic map ratings, the Food and Places tab explicitly prioritizes authentic recommendations from other international students.
- **Scale and Time Management:** The dashboard and calendar successfully centralize scattered deadlines, emails, and local events into one unified view, replacing the need to constantly check multiple university portals.

3. Calibrating Mental Models and AI Safety

Because housing agreements involve high legal stakes, we established strict interaction policies and onboarding messaging to prevent users from assuming the AI bears legal responsibility.

- **Clear Onboarding Boundaries:** Our structured onboarding variations explicitly state what the AI cannot do, such as legally signing documents, guaranteeing restaurant quality, or automatically registering the student for classes.
- **Friction and Disclaimers:** When a document is translated, the UI displays a prominent warning stating that the app is not responsible for the paperwork, the original document remains legally binding, and the user should contact the office directly for clarity.
- **Authority Calibration:** To ensure users do not assume a conversational AI has the authority to negotiate with dormitory staff, the system alters its behavior during the registration workflow to use a strictly formal and objective tone.

4. Driving Retention with Gamification

We identified that user goals shift significantly after the first two weeks as they transition from User Group A to User Group B. To solve the potential drop off in engagement once the initial administrative panic subsides, we implemented a specific gamification element. We want to keep users checking the app daily, so we integrated a fully functioning Wordle game playable in English, Polish, and Slovak. Offering this daily challenge encourages users to log in every day to maintain their streaks. This consistent daily engagement naturally increases their ongoing exposure to the calendar and social discovery features of the platform.

Mental Models + Expectations

Alex is our persona used as an example of a newcoming Erasmus student.

Part 1: Creating Onboarding

1. Onboarding templates:

- **Version 1 (Focus on Administration & Housing):** This is your Roamly Digital Student Guide, and it'll help you by centralizing university communication and translating official housing documents into your native language. It's NOT able to legally sign documents for you; you still must input your student ID, arrival date, and physical signature. Over time, it'll change to become more relevant to you. You can help it get better by interacting with the real-time validation to ensure your forms meet administration standards.
- **Version 2 (Focus on Social & Local Life):** This is your Roamly Platform Assistant, and it'll help you by finding affordable food places and calm, (non)alcoholic student events. It's NOT able to guarantee a restaurant's quality beyond international student recommendations, as generic map ratings vary by country. Over time, it'll change to become more relevant to you. You can help it get better by prioritizing specific variables when you search for nearby options.
- **Version 3 (Focus on Time Management):** This is your Roamly Digital Assistant, and it'll help you by scanning scattered emails, portal announcements, and complex university websites to extract relevant dates into a single, unified calendar. It's NOT able to register you for classes or mandatory orientations automatically. Over time, it'll change to become more relevant to you. You can help it get better by confirming the cultural events and deadlines you want added to your schedule.

2. Messaging checklist

- - **Ideas, action items, or experiments that can improve your messaging:**
Action Item: Ensure the messaging focuses on the benefit of reducing Alex's "immense mental load," rather than just listing features like "AI Translation UI".
 - *Experiment:* Test "inboarding" the translation feature. Instead of explaining it upfront, wait until Alex views the Official Registration Form and clicks the "Need Help" path before overlaying native language labels.
 - *Action Item:* Make it explicitly clear that the user must *task* the assistant to scan communications; it does not automatically read their emails without permission.

3. Demonstrating cause and effect

- - **Outline what actions the user will do next in order to reinforce the information:**
Alex opens the app and clicks the welcome notification to "Start Dormitory Check-In".
 - The system fetches the official local form.
 - Alex evaluates if they understand the local language. If "No", they trigger the AI translation, and the app overlays native language labels on the fields.
 - Alex manually fills in personal details (student ID).
 - The AI highlights missing fields, demonstrating real-time validation.
 - The AI generates correctly filled documents, so Alex can copy it onto real, physical form in the dormitory.

Part 2: Existing vs. New Mental Models

1. Defining our user groups:

- - **User group A:** New students on Erasmus exchange program
Who falls into this group? Students on Erasmus exchange program, planning on living in a university dormitory, and in the first one to two weeks of their program.
What is the primary goal of this user group? To successfully navigate the first 24 hours (getting to the dorm, understanding basic info) and handle administrative procedures without being overwhelmed by the language barrier.
 - **User group B:** Existing students on Erasmus exchange program
Who falls into this group? Students on Erasmus exchange program, after the first two weeks of their program. Searching for activities or places of interest.
What is the primary goal of this user group? To successfully navigate the student life

2. Mapping novice and expert workflows

- **User group A (Novice/Actual workflow):**

- **Process:** Constantly switching between apps like Google Maps, WhatsApp, and ChatGPT to translate information. Using Google to translate official forms and manually asking friends or staff for help. Spending significant time looking through dozens of Google Maps results.
- **Uniformity:** Highly variable (relies heavily on trial, error, and scattered tools).
- **User group A (Expert/Intended AI workflow):**
 - **Process:** Tasking the digital assistant to scan centralized communications to extract and translate all relevant dates into a single calendar. Relying on the platform to translate official housing documents directly within the UI.
 - **Uniformity:** Highly uniform (centralized within one platform).
- **User group B (Novice/Actual workflow):**
 - **Process:** Constantly switching between apps like Google Maps, WhatsApp, and ChatGPT to get and translate information. Looking through the WhatsApp group in search of activities. Spending significant time looking through dozens of Google Maps results.
 - **Uniformity:** Highly variable (relies heavily on trial, error, and scattered tools).
- **User group B (Expert/Intended AI workflow):**
 - **Process:** Tasking the digital assistant to scan centralized communications to extract and translate all relevant dates into a single calendar. With notifications of upcoming events
 - **Uniformity:** Highly uniform (centralized within one platform).

3. Analyze existing mental models

- **What mental models might already be in place?** Alex does not trust generic map ratings, knowing they mean completely different things in different countries. They also expect a lack of specific information regarding local health systems and cost of living prior to arrival.
- **Where could the user's mental model break?** The model might break if the AI recommends a restaurant based strictly on high star ratings rather than the specific variable of "authentic recommendations from other international students," leading Alex to distrust the tool.
- **What cause and effect relationships does the user need to understand?** Alex needs to understand that if they lack information on an official form, the AI will

highlight missing fields, but it will *not* automatically submit the form without their manual input of sensitive IDs. And they need to be aware that AI is only checking their documents and just like a human can make mistakes.

- **How might adding human-like cues alter the mental model?** If the AI is too conversational or "human-like" during the housing registration, Alex might assume the AI has the authority to directly contact Dormitory Support and negotiate on their behalf, rather than just acting as a translation overlay.

4. Calibrate user mental models

- **The biggest risks to users developing inaccurate mental models are:** Believing the application entirely removes their personal legal responsibility when signing dormitory paperwork, or assuming the calendar automatically registers them for classes rather than just tracking deadlines.
- **List the key points in a product where messaging is critical:**
 - **"Onboarding":** When Alex first opens the digital companion upon arrival to view the high-level checklist of urgent day-one tasks.
 - **"Inboarding":** When Alex encounters the "Understand local language?" decision node during dormitory check-in, requiring clear messaging on what the AI Translation UI actually does.

5. What, if anything, might need to change about how the AI works to accommodate mental models?

- **All users (Alex):** The AI's food recommendation algorithm *must* prioritize the "authentic recommendations from other international students" data point over standard API map ratings to bypass Alex's existing negative mental model of map reviews.
- **All users (Alex):** Because users might assume a conversational AI has the authority to negotiate with dormitory staff, the AI system must alter its behavior during the Registration Workflow. It must switch to a strictly formal, objective tone and disable conversational small talk to reinforce that it is merely a translation overlay and not a legal representative.

!Putting disclaimer that the app is not the responsible for the paperwork

More clarity for the user

1. Evidence of User Need ②

Worksheet: User research summary

- **Date:** Recent.
- **Source:** Persona Creation From Survey Results.
- **Summary of findings:** Alex is a 21-year-old third-year bachelor exchange student who just arrived in their host city. They are dealing with an immense mental load from administrative procedures, securing dormitory accommodation, and navigating a foreign language. They rely heavily on their mobile phone, constantly switching between apps to translate information and find their way around.

1. Statement of User Need ②

- **How might we solve:** the immense mental load and language barrier faced by new exchange students trying to navigate official administrative procedures, housing registration, and local discovery? / Repetitive task of translating documents, searching for events.
- **How pervasive is the problem / Whom does it affect:** It affects typical exchange students who have just arrived in a new host city and rely on their mobile phones to navigate daily life. / It affects all of the students on the Erasmus exchange.
- **How does this problem vary across dimensions:** It is particularly severe for students who struggle significantly with the language barrier when communicating with locals or handling official paperwork.
- **Do the problem's effects change over time:** Yes, it is most critical during the first one to two weeks of their program (first day arrival and orientation). / Then the user transitions to the second group.

2. AI Suitability (What to check)

We checked boxes:

- **"Language":** The core user experience requires natural language interactions. Alex struggles significantly with the language barrier when communicating with locals or handling official paperwork and relies on the platform to translate official housing documents into their native language.
- **"Personalization":** The core experience requires personalizing content to individual users. Alex wants to easily find affordable food places based on specific variables (like authentic recommendations from other international students) rather than trusting generic map ratings.

- **"Scale"**: You need to recognize a general class of things that is too large to articulate every case. Alex is overwhelmed by scattered emails, portal announcements, and complex university websites and spends significant time looking through dozens of Google Maps results. The AI addresses this by scanning centralized university communications to extract relevant dates.

3. Summary of Evidence

- **We think AI CAN help solve** the user's need.

Because: The core experience requires natural language interactions to translate official documents and forms. Furthermore, the user experience is significantly enhanced by personalization, as Alex wants to easily find affordable food places based on specific variables (authentic recommendations from international students) rather than generic map ratings.